Anirudh Kowshik

Systems Validation Engineer, AI Infrastructure

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SUMMARY

Systems validation engineer in AI infrastructure with **5+ years** across the stack: validating **NVIDIA GB300/VR200** and Meta OCP AI racks from bring-up to mass production, standing up bare-metal **Kubernetes** and **Slurm** compute, and shipping **LLM/RAG** systems. Debugs from SerDes links and BMC firmware up to RDMA/RoCE fabrics and cluster software, and automates repetitive validation in Python and Bash.

SKILLS

AI hardware & validation: NVIDIA GB300/VR200 racks, OCP servers, NPI (EVT/DVT/PVT), ASIC-based rack validation, L10–L12 bring-up, PXE, PRBS/SerDes, PCIe, NVMe, BIOS, CPLD, BMC (Redfish, IPMI, SOL)

Networking: NVLink, NVSwitch, NVIDIA Fabric Manager, RDMA, RoCE, NCCL

Cloud & platform: Bare-metal provisioning, Kubernetes (VMware Tanzu), Slurm, Terraform, Ansible, Prometheus, Grafana, Docker, Trivy, CI/CD (Azure DevOps, Jenkins, GitHub Actions), AWS, Azure, GCP

Programming & ML: Python, Bash, SQL, PyTorch, NumPy, pandas, scikit-learn, XGBoost, LangChain, RAG, LLMs, Kafka, Linux

EXPERIENCE

Systems Validation Engineer — Quanta Manufacturing, Fremont, CA — Jan 2026 – Present

- Took **NVIDIA GB300 and VR200** AI racks and Meta OCP server platforms through NPI toward mass production, gating each build with test plans, firmware validation, failure debug and fixes driven back to hardware and firmware teams.
- Validated custom ASIC-based AI racks across **4 NPI builds** (**EVT, DVT, PVT**) toward **mass production**, using **PXE** network boot to provision multi-host test environments for **L10–L12** rack bring-up.
- Verified **RDMA/RoCE** networking across NICs, network modules and the full rack topology on every build, catching **35+** link, connectivity and configuration issues at rack level before they could reach production AI clusters.
- Root-caused **20+** PCIe, NIC, NVMe and retimer link failures with **PRBS/SerDes** signal-integrity testing, isolating issues across hardware, firmware and test infrastructure so fixes landed in the next board rev or firmware RC.
- Hardened **BMC firmware** across **12 release candidates** by validating Redfish, IPMI, SOL and out-of-band management end to end, including sensors, telemetry and BMC–BIOS/CPLD/BIC interactions on production-intent hardware.
- Replaced repetitive manual checks with **Python/Bash automation**, a hardened Python/YAML test framework and log-parsing tooling, automating **~60%** of manual rack checks and saving roughly **two engineer-days per build**.
- Debugged **NVLink** switch fabric issues with NVIDIA **Fabric Manager**, tracing GPUs that failed to join the fabric back to link, firmware or configuration faults so each rack came up with all **72 GPUs** on the NVLink fabric.

Data Scientist — ResMed, Remote — Oct 2024 – Dec 2025

- Launched **Dawn**, a production **RAG** patient-support assistant built on LangChain, **Amazon Bedrock** Titan embeddings and OpenSearch Serverless, giving patients safe, context-aware answers grounded in retrieved support content.
- Cut LLM hallucinations by **40%** by designing system prompts and safety guardrails, keeping patient-facing answers compliant.
- Lifted marketing conversion **17%** and lowered customer acquisition cost **12%** by designing and analyzing A/B tests.
- Built **ARIMA, Prophet and XGBoost** demand forecasts in Python (**PyTorch**, NumPy, pandas, scikit-learn) on historical sales and patient-adoption data, giving the business forecasts that enabled strategic production planning.

Data Engineer — DLL, Wayne, PA — Jun 2023 – Jun 2024

- Migrated **5M+** insurance records from Oracle to **Azure Data Lake Gen2** with real-time **Kafka/MirrorMaker** replication, giving analysts cloud-native access to data that previously lived only in Oracle.
- Sped up downstream Power BI reporting by tuning complex SQL in **Azure Data Factory** pipelines feeding the dashboards.
- Authored team-wide SQL and ETL coding standards that were adopted across **5 projects**, raising peer-review code quality.

Software Engineer — Unisys, Remote (India) — Sep 2020 – Jul 2022

- Provisioned **30+ on-prem bare-metal nodes** with **Terraform** and **Ansible** playbooks, then stood up **3 Tanzu Kubernetes clusters** on top to give **2 engineering teams** their staging environments.
- Built **Prometheus/Grafana** monitoring across the bare-metal clusters for node metrics, pod storage alerts and cluster resource usage.
- Managed **20+ Azure VMs** for heavy data processing with **Terraform** modules and **Ansible**-automated patching.
- Wrote the **Slurm** job templates and sbatch submission scripts for batch workloads, sizing memory per job so runs completed without out-of-memory kills, and tracked queue depth and node utilization in **Grafana**.
- Cut release-cycle effort **20%** by building automated **Trivy** vulnerability scanning into the Azure DevOps release pipeline.

EDUCATION

PhD, Information Technology (Computer Engineering), executive — University of the Cumberlands — Expected 2030

MS, Data Analytics · Chancellor's Scholar — Penn State University

BE, Computer Science — The National Institute of Engineering, Mysore